

Qatar-4b

R-2607

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-4_201008 · created 2026-08-28T01:10:50+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459130.8076 +/- 0.0022 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.12 +/- 0.047
Transit depth (Rp/Rs)^2	1.43 +/- 1.13 [%]
Orbital Inclination (inc)	85.05 +/- 2.08
Airmass coefficient 1 (a1)	0.924 +/- 0.027
Airmass coefficient 2 (a2)	0.065 +/- 0.069
Scatter in the residuals of the lightcurve fit is	2.97 %
Best Comparison Star	#3 - [322, 309]
Optimal Aperture	3.06
Optimal Annulus	13.76
Transit Duration (day)	0.071 +/- 0.019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.12 $\pm$ 0.047 vs 0.138 (deviation 0.38 σ)
Duration fit vs published	0.071 d vs 0.0892 d (deviation 0.96 σ)
Mid-time O-C	-5.1 min
Depth SNR	2.6
Residual scatter	2.97 %

Light curve

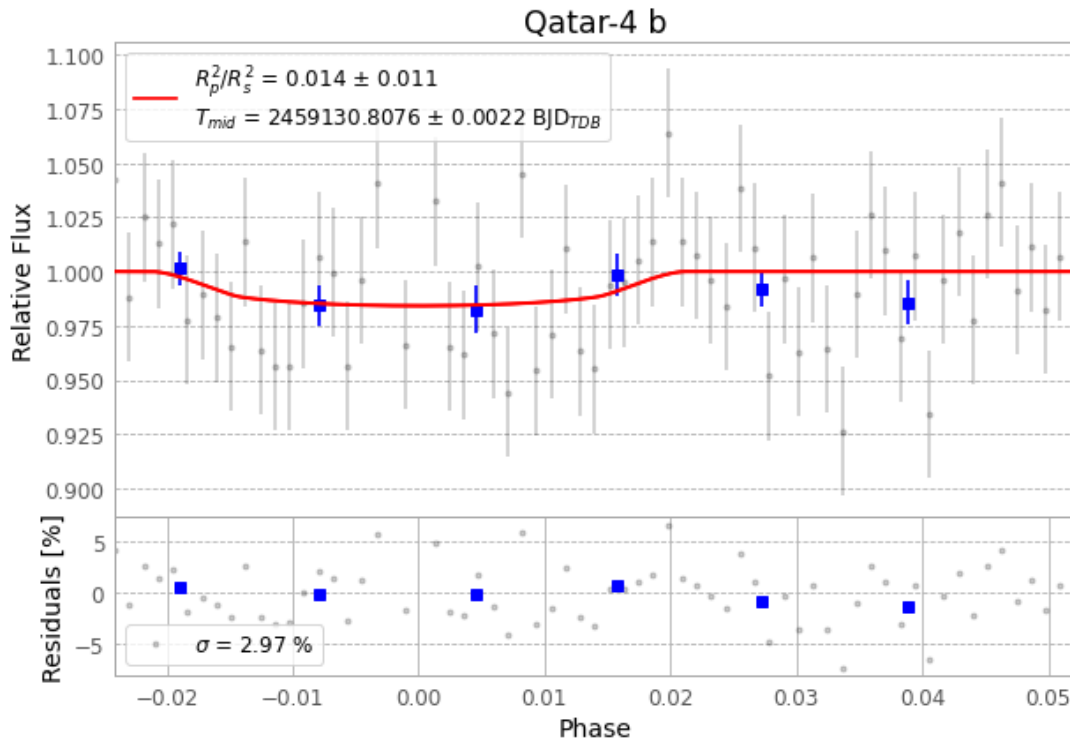


Figure 1 — FinalLightCurve_Qatar-4 b_2020-10-07.png (SHA-256 8df7fd7677ea80b9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [363, 328], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 54755eb7d11bc926...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

67 FITS frames (44 MB), Qatar-4201008061212.FITS → Qatar-4201008093012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-4-201008.log (9c35738f979f...), FinalLightCurve_Qatar-4 b_2020-10-07.png (8df7fd7677ea...), FinalLightCurve_Qatar-4 b_2020-10-07.csv (30543359a3ae...)

Compute resources

Wall time 107.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-28 01:10Z.